
DSAA2011 Covertypes Project Report

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DSAA2011 Machine Learning Course Project
groupID_covertypes

Abstract

This report studies the UCI Covertypes dataset with 581,012 forest observations, 54 cartographic features, and seven cover-type classes. The analysis covers preprocessing, t-SNE visualization, MiniBatchKMeans and Ward clustering, supervised classification, model validation, a Wilderness Area extension, feature importance, calibration, robustness, and computational optimization. Natural clusters reveal terrain and region structure but do not recover the seven ecological labels. A Random Forest classifier is the strongest predictive model on the stratified test set, achieving accuracy=0.902, macro-F1=0.845, and macro one-vs-rest AUC=0.991.

1 Introduction

The Covertypes task predicts forest cover type from cartographic variables. The target is imbalanced: Lodgepole Pine and Spruce/Fir account for most observations, while Cottonwood/Willow is only 0.5% of the data. We therefore report macro-averaged metrics in addition to accuracy. The main empirical result is that unsupervised structure is visible but does not directly match the supervised label definition, whereas nonlinear supervised ensembles perform well.

Table 1: Class distribution in the Covertypes dataset.

Cover_Type	Cover_Type_Name	count	proportion
1	Spruce/Fir	211,840	36.5%
2	Lodgepole Pine	283,301	48.8%
3	Ponderosa Pine	35,754	6.2%
4	Cottonwood/Willow	2,747	0.5%
5	Aspen	9,493	1.6%
6	Douglas-fir	17,367	3.0%
7	Krummholz	20,510	3.5%

2 Data preprocessing

The raw UCI file is numeric and has no header. We assign the official feature names, use 10 continuous terrain measurements, 4 wilderness-area indicators, 40 soil-type indicators, and the seven-class Cover_Type target. There are no missing values. Continuous features are standardized for linear and distance-based methods; binary indicators are kept as indicator variables. The implemented data optimization uses float32 for continuous variables and uint8 for indicators and target, reducing the naive all-int64 memory estimate from 243.8 MB to 47.1 MB.

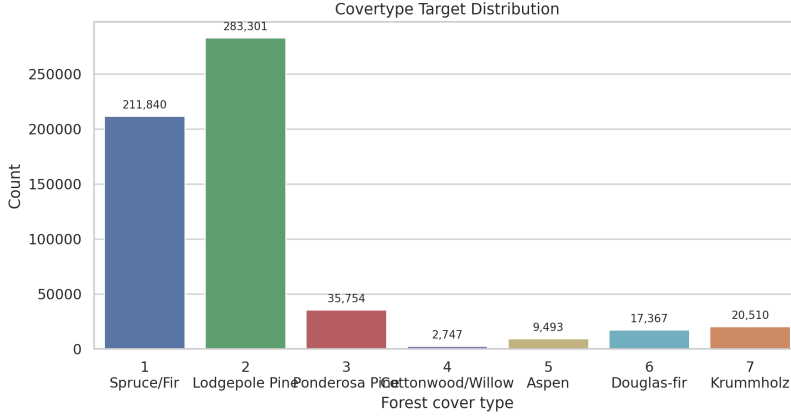


Figure 1: Cover-type distribution. The two dominant classes make macro metrics necessary.

3 Visualization and clustering

t-SNE is applied to a stratified 6,000-row sample after standardization and PCA compression to 30 dimensions. The embedding shows partial local separation for high-elevation classes, but Spruce/Fir and Lodgepole Pine overlap substantially. This motivates supervised classification rather than relying on natural clusters.

MiniBatchKMeans and Ward hierarchical clustering are evaluated with silhouette, Calinski-Harabasz, adjusted Rand index against *Cover_Type*, and adjusted Rand index against *Wilderness_Area*. MiniBatchKMeans is used as the scalable K-Means variant; Ward is restricted to a smaller structural sample because of its quadratic memory/time behavior. The near-zero ARI against *Cover_Type* should not be read as a failure to find any structure. It shows that the strongest unsupervised partitions are not equivalent to the seven ecological labels; the clusters are better interpreted as coarse terrain or regional groupings.

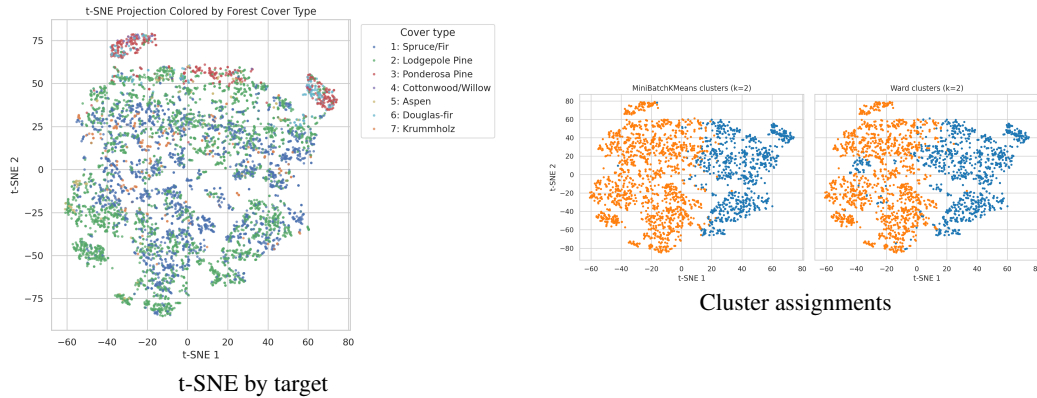


Figure 2: Low-dimensional visualization and clustering results.

Table 2: Best clustering results under the selected internal metrics.

algorithm	k	silhouette	CH	ARI target	ARI wild
MiniBatchKMeans	2	0.191	1340.066	0.003	0.039
Agglomerative Ward	2	0.153	406.143	0.004	0.013

4 Wilderness Area extension

The four one-hot wilderness indicators are converted into a compact `Wilderness_Area` variable. This gives an additional explanatory view that is distinct from the seven-class prediction target. The areas differ strongly in size and cover-type composition: each wilderness area has a dominant cover type, but the normalized entropy shows that some areas contain more mixed ecological labels than others. This supports the revised clustering interpretation: low ARI versus `Cover_Type` only says that natural clusters are not the target labels; it does not rule out meaningful structure associated with region, elevation, and terrain.

Table 3: Cover-type composition summary by wilderness area.

area	name	n	share	dominant cover	dominant share	entropy
1	Rawah	260,796	44.9%	Lodgepole Pine	56.1%	0.426
2	Neota	29,884	5.1%	Spruce/Fir	62.2%	0.439
3	Comanche Peak	253,364	43.6%	Lodgepole Pine	49.4%	0.628
4	Cache la Poudre	36,968	6.4%	Ponderosa Pine	58.0%	0.547

The same held-out predictions from the best supervised model are then grouped by wilderness area. Accuracy and macro-F1 vary across regions, so model quality is not spatially uniform. The regional split is therefore useful as a robustness check, not just as four extra binary predictors.

Table 4: Best-model test performance by wilderness area.

area	n	test share	accuracy	macro-F1	dominant cover
Rawah	16,228	0.451	0.905	0.846	Lodgepole Pine
Neota	1,831	0.051	0.944	0.943	Spruce/Fir
Comanche Peak	15,657	0.435	0.903	0.848	Lodgepole Pine
Cache la Poudre	2,284	0.063	0.842	0.771	Ponderosa Pine

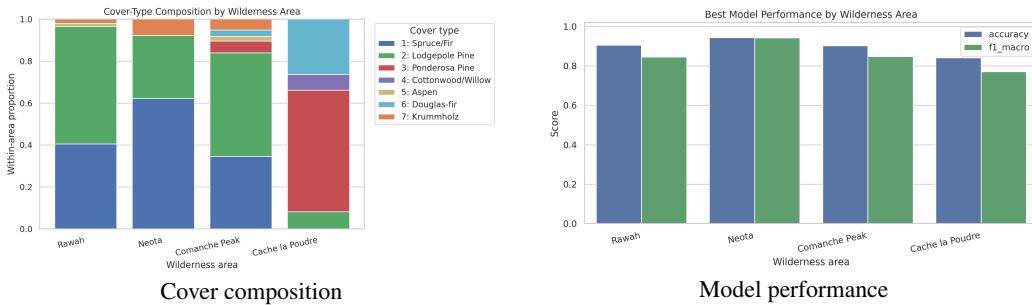


Figure 3: Wilderness-area extension analysis.

5 Prediction: training and testing

The supervised target is the seven-class `Cover_Type`. We use a stratified 120,000-row modeling sample, split 70/30 into 84,000 training and 36,000 test observations. Logistic regression is the linear baseline with class weighting, and a depth-limited decision tree is the simple nonlinear baseline. Both are evaluated on train, test, and full data after training.

Table 5: Simple-model performance on train, test, and full data.

model	split	accuracy	f1_macro	AUC_ovr_macro
Logistic Regression	train	0.594	0.473	0.923
Logistic Regression	test	0.593	0.471	0.923
Logistic Regression	full	0.593	0.472	0.923
Decision Tree	train	0.826	0.727	0.985
Decision Tree	test	0.775	0.663	0.942
Decision Tree	full	0.782	0.671	0.949

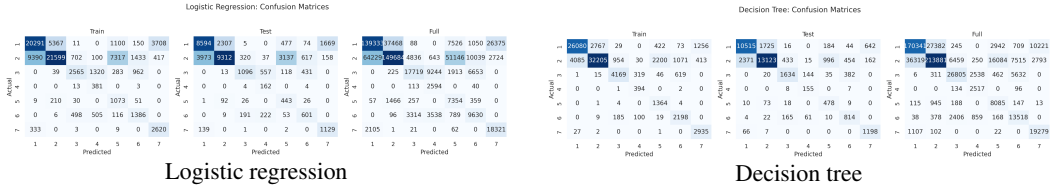


Figure 4: Confusion matrices for the two required simple models.

6 Evaluation and model choice

Logistic regression reaches high macro AUC but weak macro-F1 because the linear boundary struggles with minority classes. The decision tree improves nonlinear separation but still overfits relative to the test set. Random Forest and HistGradientBoosting are added for open-ended model comparison. Random Forest is the best test model with accuracy=0.902, macro-F1=0.845, and macro AUC=0.991. Cross-validation on a separate stratified 36,000-row sample confirms the same ranking.

Table 6: Test performance across model classes.

model	accuracy	precision_macro	recall_macro	f1_macro	AUC_ovr_macro
Logistic Regression	0.593	0.447	0.690	0.471	0.923
Decision Tree	0.775	0.596	0.816	0.663	0.942
Random Forest	0.902	0.853	0.840	0.845	0.991
HistGradientBoosting	0.869	0.866	0.806	0.831	0.985

Table 7: Three-fold cross-validation on a stratified sample.

model	accuracy_mean	f1_macro_mean	AUC_ovr_macro_mean
Random Forest	0.844	0.755	0.978
HistGradientBoosting	0.826	0.739	0.973
Decision Tree	0.693	0.569	0.907
Logistic Regression	0.598	0.496	0.925

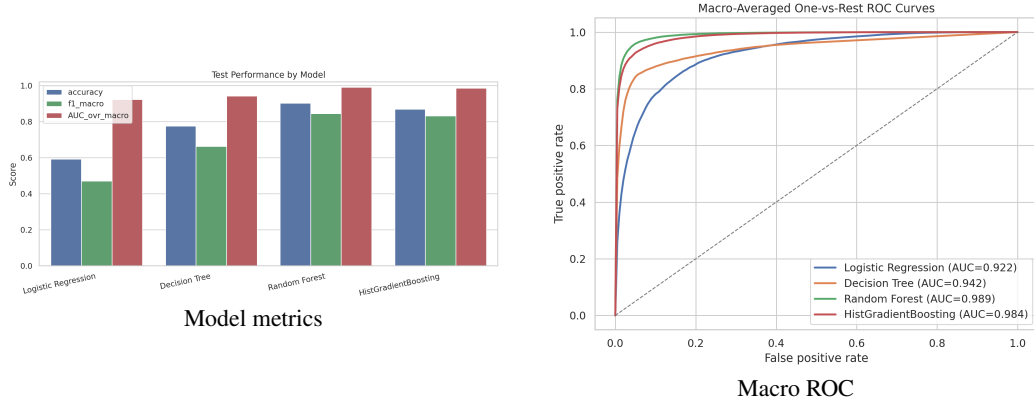


Figure 5: Model comparison and macro one-vs-rest ROC curves.

7 Open-ended exploration

7.1 Feature importance and ablation

Random Forest impurity importance and sampled permutation importance both rank Elevation as the dominant feature, followed by road/fire/hydrology distances and hillshade. Feature-group ablation with HistGradientBoosting shows that topographic variables alone are already strong, but soil indicators add useful predictive signal; wilderness indicators alone are weak.

Table 8: Top Random Forest impurity importances.

feature	importance
Elevation	0.235
Horizontal_Distance_To_Roadways	0.099
Horizontal_Distance_To_Fire_Points	0.077
Horizontal_Distance_To_Hydrology	0.062
Vertical_Distance_To_Hydrology	0.050
Hillshade_9am	0.049
Aspect	0.044
Hillshade_3pm	0.042

Table 9: Feature-group ablation using HistGradientBoosting.

experiment	n_features	accuracy	f1_macro	AUC_ovr_macro
All features	54	0.857	0.815	0.983
Topographic + soil	50	0.850	0.807	0.981
Topographic + wilderness	14	0.837	0.784	0.973
Topographic only	10	0.820	0.764	0.973
Soil only	40	0.618	0.279	0.873
Wilderness only	4	0.535	0.203	0.741

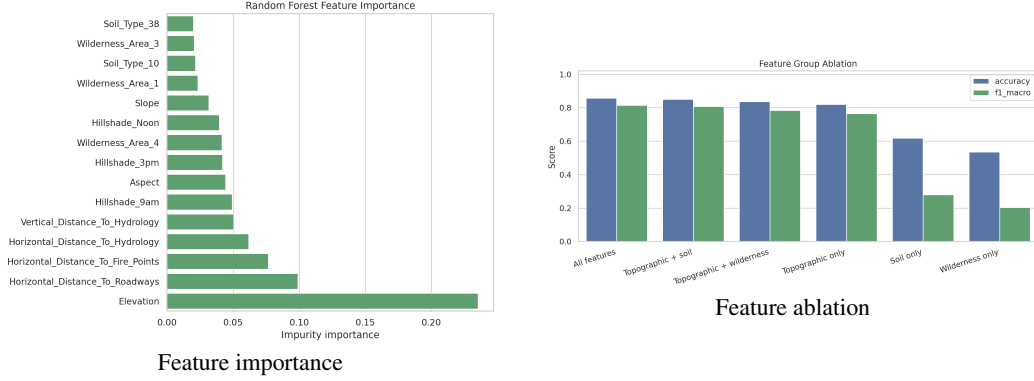


Figure 6: Feature-importance and feature-group analyses.

7.2 Calibration, robustness, and errors

HistGradientBoosting has the lowest top-label expected calibration error, while Random Forest has the best accuracy and macro-F1 but is under-confident on average. Accuracy also rises in the highest elevation band, indicating that high-elevation classes are easier to separate. Class-level recall shows Aspen and Douglas-fir are harder than the dominant classes.

Table 10: Top-label calibration summary.

model	multiclass_brier	expected_calibration_error	mean_confidence	observed_accuracy
HistGradientBoosting	0.194	0.047	0.822	0.869
Logistic Regression	0.546	0.066	0.655	0.593
Decision Tree	0.324	0.104	0.880	0.775
Random Forest	0.178	0.128	0.775	0.902

Table 11: Class-level precision, recall, and F1 for the best model.

Cover_Type	Cover_Type_Name	support	precision	recall	f1
1	Spruce/Fir	13126	0.919	0.885	0.901
2	Lodgepole Pine	17554	0.907	0.930	0.918
3	Ponderosa Pine	2215	0.852	0.924	0.887
4	Cottonwood/Willow	170	0.782	0.824	0.802
5	Aspen	588	0.792	0.639	0.707
6	Douglas-fir	1076	0.796	0.778	0.787
7	Krummholz	1271	0.920	0.901	0.911

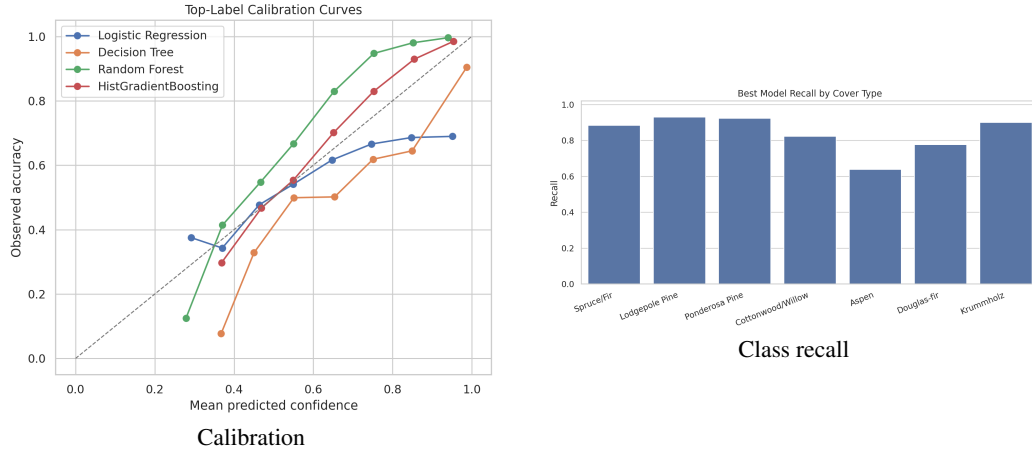


Figure 7: Calibration and class-level recall diagnostics.

7.3 Computational optimization

The optimized workflow implements the cost controls discussed in the project: dtype downcasting, stratified samples for expensive diagnostics, PCA before distance-based analysis, MiniBatchK-Means for scalable clustering, parallel tree ensembles, and manifest-based caches for t-SNE, clustering, and modeling outputs. A warm rebuild hits the heavy-stage caches, making report regeneration fast and auditable.

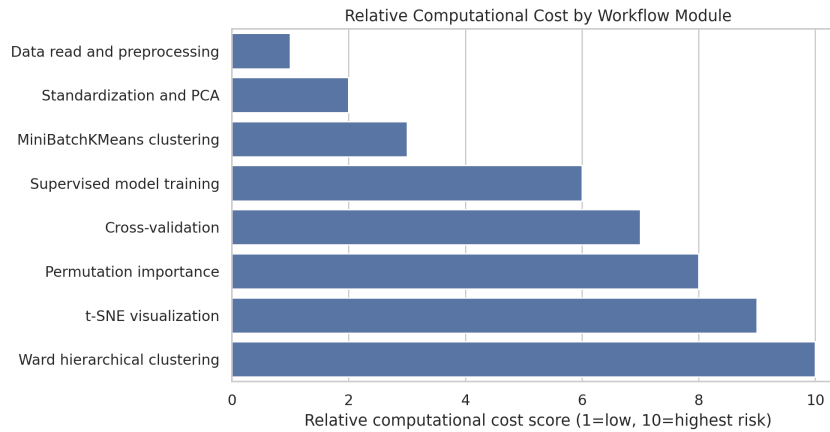


Figure 8: Relative computational cost by workflow module.

Table 12: Runtime profile after cache-enabled optimization.

stage	seconds	cache_status
data_read_downcast	0.88	computed
descriptive_outputs	1.00	computed
computational_cost_outputs	0.21	computed
t-SNE	0.00	cache_hit
clustering	6.89	computed
modeling	135.25	computed

8 Conclusion

The Covertypes dataset has strong class imbalance and overlapping ecological classes. Unsupervised clustering exposes broad terrain and regional structure, but the near-zero ARI against `Cover_Type` only shows that these natural partitions are not the same object as the seven supervised labels. The Wilderness Area extension makes this distinction explicit by comparing cover-type composition and held-out model performance across regions. Supervised tree ensembles are substantially more effective, with Random Forest giving the best overall classification performance and HistGradientBoosting offering better calibration. Elevation is the strongest feature, but road/fire/hydrology distances, soil context, and regional evaluation all add meaningful interpretation. For deployment-style interpretation, the model should be reported with macro metrics, class-level recall, calibration, terrain-band and wilderness-area performance, and explicit sampling assumptions.

References

1. UCI Machine Learning Repository. Covertypes dataset. <https://archive.ics.uci.edu/ml/datasets/covertypes>.
2. Blackard, Jock A. and Dean, Denis J. Comparative Accuracies of Artificial Neural Networks and Discriminant Analysis in Predicting Forest Cover Types from Cartographic Variables. *Computers and Electronics in Agriculture*, 1999.
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Credit and GenAI use

Analysis code, figures, and report text were generated reproducibly in this repository. Group-member contribution details should be finalized by the project team before submission. OpenAI Codex was used to draft, refactor, and validate the analysis workflow and report.